Beyond discrimination: toward trustworthy autism classification from connectomes

Cohort and method

871 participants
20 sites, ABIDE-I

Three views
of connectivity

Training-only
logistic models

Discrimination (AUROC)

0.776	0.774
ten-fold	leave-one-site-out

Five findings

1. unmatched IQ bands — **AUROC 0.873 → 0.719 as IQ rises**
   no within-autism symptom link after correction
2. all matched — **Somatomotor reduction strengthens**
   after within-site motion matching (exploratory)
3. linear trees — **Linear 0.735–0.757 vs trees 0.611–0.624**
   AUROC; windowing gain gone with matched shrinkage
4. native recalibrated — **Offset −0.162 → −0.010, AUROC unchanged**
   site offsets vary with autistic proportion
5. system pairs connections — **Default-mode pairs lead; patterns reproduce**
   site-disjoint halves ρ 0.54–0.63; chance ≤ 0.10

Conclusion: reliable connectome-based autism classification requires evaluating discrimination, calibration, cohort characteristics, analytical design, and explanation reproducibility.